

Optimal Reservoir Operations to Maximize Profit During Irrigation Season

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Introduction

During the irrigation season, prices increase as the season gets closer to the end. The reservoir has a finite storage, meaning the irrigator can't wait until the last minute to distribute all the water. The report explains how to operate the reservoir to maximize profit. This report gives the model formulation, the optimal solution, and a model analysis.

Model Formulation

The dimension of the model are the months (June, July, and August) and location of water (Spillway, Diversion, and Storage). The decision variables are how much water should go to each location during each month. These variables are expressed as $X(\text{location}, \text{month})$ in units of acre-ft. This notation means there are nine decision variables.

The objective function maximizes profit based on the irrigation benefits (Table 1). The irrigations benefits are expressed as $c(m)$. The objective function can be expressed as $\text{Max}(Z) = \sum c(m_i) * X(\text{Diversion}, m_i), [\$]$. The other decision variables are only not included in the objective function because they do not contribute any profit.

The constraints are listed with the corresponding mathematical equation and units: The ending storage is greater than or equal to the beginning storage, $X(\text{Storage}, \text{Aug}) \geq 3000, [ac - ft]$.

The max capacity of the reservoir is 9000 ac-ft, $X(\text{Storage}, m_i) \leq 9000, [ac - ft]$. The reservoir must follow the mass balance equation. $X(\text{Stor.}, m_i) - X(\text{Stor.}, m_{i-1}) = \text{inflow}_i - X(\text{Div.}, m_i) - X(\text{Spill}, m_i), [ac - ft]$. (Note: inflows are shown in Table 2)

Solution

GAMS (GAMS Development Corporation, 2026) found the optimal solution that provides the maximum profit. Table 3 contains the optimal diversion schedule. The maximum profit, given by the objective function, is \$3,774,000. Based on the GAMS solution the constraints that binds the objective function are the mass balance constraint during all months, the ending storage being greater than or equal to initial storage in August, and the reservoir capacity in July. The rest of the constraints are slack during the respective months.

Analysis

The diversion schedule is optimal because the most water diverted is in the last month and none of the water goes on the spillway to the river. All water contributes the most profit possible. This problem helped grow my understanding on model formulation; specifically, how to mathematically express the model. This model does not consider the demand of farmers which might be a concern for the company manager.

Conclusion

In conclusion by following the optimal diversion schedule the profit will be \$3,774,000. This is the maximum profit based on the constraints and information given. GAMS programing and modeling software was used to find the solution to the problem.

Appendix A – Tables

Table 1: Irrigation Benefits (Rosenberg, 2026)

Month	Irrigation Benefits (\$/acre-foot)
June	150
July	170
August	425

Table 2: Inflows (Rosenberg, 2026)

Month	Inflows (acre-feet)
June	5,000
July	3,200
August	2,000

Table 3: Diversion Schedule

Month	Diversion Amount (acre-feet)	Corresponding Reservoir Level (acre-feet)
June	0	8,000
July	2,200	9,000
August	8,000	3,000

Appendix B - References

GAMS Development Corporation. (2026). General Algebraic Modeling System (GAMS) (Release 54.4.0) [Computer software]. <https://www.gams.com/>

Rosenberg, D. (2026). HW3-LinearPrograminGAMS. Logan; Utah State University.

AI Statement

I used AI (Gemini) to help me fix some errors in my GAMS. I was having a hard time using the \$ operator in GAMS and I couldn't figure out the syntax. AI helped me understand how the syntax works. This follows the AI policy because I did not copy and paste anything from AI into my GAMS. I did not let AI govern any decision I made in my model. Here is the AI chat:

<https://share.gemini.google/kPXsfVZF182a>

Category (Max. Score)	No Evidenc e	Doesn't Meet Standard	Nearly Meets Standard	Meets Standard	Exceeds Standard	Self- Score	Instructo r Score
Title (1)	Absent 0	Evidence of two or less 0	Evidence of three 0	Evidence of four 1	Title – can assess main point from title alone; Name, Instructors' Names, Course, Date, Neatly finished 1	1	
Introduction (3)	Absent, no evidenc e 0	There is no clear introduction or main topic. 1	Introduction states the main topic but either: 1. Does not give a full overview, Or: 2. Too detailed, leading to annoying repetition later. 2	The introduction states the main topic and previews the structure of the report. 2	The introduction states the main topic and previews the structure of the report. Good overview of the problem and solution approach. Gives enough detail to motivate the reader to continue reading. 3	3	
Organization and structural developmen t of the idea: procedure, results, conclusions (10)	No content provided. 0	Paragraphs fail to develop the main idea. No section headers or guide to help the reader understand how material is organized. 1 – 5	Organization of ideas not fully developed. Paragraphs lack supporting detail sentences. No transitions and/or ineffective section headers. 6 - 7	Paragraph development present but not perfected. Each paragraph has sufficient supporting detail sentences. Few transitions. 8	Writer demonstrates logic and sequencing of ideas through well-developed section headers, paragraphs, and transitions. The first sentence of each paragraph is the summary sentence. 9 - 10	9	
Technical Correctness	Questions not addressed.	The writer has no clue what they are talking	Sketchy: left out required design points. Did not work	Discussion lacks adequate detail, but all the	Provides what was explicitly asked for. The function of each piece is		

(70)	3 – 42%	about. 45 – 58%	on this as much as you should have, and it shows. Many important answers are incorrect. 61 – 79%	necessary points are covered and nearly all answers are correct. 82 – 88%	demonstrated to the reader in adequate, but not overwhelming, detail. Answers are correct and reasonable. 91 – 100%		
	a) Problem description (10)					10	
	b) Model Formulation – Dimensions (5), Decision Variables (10)					15	
	c) Objective(s) (10), Constraints (10)					20	
	d) Solution - Delivery schedule, Objective Function value, Constraints that bind and that are slack (15_					15	
	e) Explain why the optimal solution is optimal, new insights you took away about the problem, and concerns the company manager will have about the model formulation and solution. (10)					10	

Category (Max. Score)	No Evidenc e	Doesn't Meet Standard	Nearly Meets Standard	Meets Standard	Exceeds Standard	Self- Score	Instructo r Score
Word Usage and Format (7)	Not applicable	Numerous and distracting errors in punctuation, capitalization, spelling, sentence structure, word usage, significant figures, tables, and figures. Data vomited onto page(s).	Misspelled words, poor English grammar and word choice. Main body of report is either longer or significantly less than one page. Figures are too small and/or under-labeled, although they are usually of acceptable quality and focus. Tables incoherent or	Almost no errors in punctuation, capitalization, spelling, sentence structure, word usage, significant figures, and presentation of figures, tables, and appendices. Main body of report is one page or less	Punctuation, capitalization, spelling, sentence structure, word usage, and significant figures all correct. Main body of report is one page or less. Clear, consistent fonts. Good word processing skills. Figures have adequate contrast. Informative figure and table titles and legends. Figures have appropriate axis tick spacing, labels,		

		Unacceptable / unprofessional at the graduate level. 1 – 5	not cohesive. Bad font sizes. Too much or too little data in appendices. Could be improved by being more meticulous. 5	6	units, and legends. Table columns cohesive, labeled, and specify units. Document is stapled. Appendices, if provided, are separated by topic, and each have a title, discussion, and proper formatting and display of information 7	6	
Conclusion (4)	Absent 0	Incomplete and/or not focused. 1	The conclusion does not adequately restate the main results. 2	The conclusion restates the main results. 3	The conclusion restates the main results, and is an effective summary. 4	4	
Use of AI (3)	Absent (Return without grade)		1	See Policy on Use in Syllabus If used , list the AI tool], provide a short description of use, how use is within the course policy, and link to the transcript of student-AI interactions. If not used , write “No AI was used in the analysis or write-up of this report.” 3		3	
References (2)	Absent 0	Numerous errors, off-the-wall sources used. 0	Some errors in citing format; more sources should be cited. 1	Prior work cited with few errors. 2	All prior work and data sources are cited in the correct format with no errors. 2	2	
TOTAL (100)						98	